Aaron Slipper

Department of Mathematics, Duke University

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Academic appointment

2024–present **Phillip Griffiths Assistant Research Professor of Mathematics**
Duke University. Research mentor: Jayce Getz. Appointment through August 2027.

Education

2024 **Ph.D. in Mathematics**, University of Chicago
Advisors: Ngô Bảo Châu and Kazuya Kato.
Thesis: *Non-Abelian Fourier Transforms and Normalized Intertwining Operators over Finite Fields for General Parabolic Subgroups*.

2018 **B.A. in Mathematics**, Harvard University, *magna cum laude*
Thesis advisor: Noam Elkies.
Thesis: *Modular Magic: The Theory of Modular Forms and the Sphere Packing Problem in Dimensions 8 and 24*.

Research interests

Number theory and geometric representation theory; nonlinear Fourier transforms and the Langlands program; logarithmic geometry.

Publications and preprints

[1] *Bosonic codes from compact phase spaces*.
With D. Roberts, A. Parhizkar, V. V. Albert, and M. Hafezi. Joint first authorship with D. Roberts.
2026. arXiv:2608.31156.

[2] *Quasi-Classical Braverman–Kazhdan Intertwiners via Quiver Varieties*.
With N. Grantcharov.
2026. arXiv:2606.14637.

[3] *Geometrization of the Schrödinger Model for the Minimal Representation of an Even Orthogonal Group: The de Rham Setting*.
2026. arXiv:2604.11023.

[4] *The ρ-Fourier transform*.
With J. R. Getz, A. Gutiérrez Terradillos, F. Hosseinijafari, G. Xi, H. Yao, and A. Zhao.
2025; revised 2026. arXiv:2512.00182.

[5] *Modulation groups*.
With J. R. Getz, A. Gutiérrez Terradillos, F. Hosseinijafari, B. Hu, S. Lee, M.-H. Tomé, H. Yao, and A. Zhao.
2025. arXiv:2510.23932.

[6] *Logarithmic Geometry and Geometric Class Field Theory*.
2025; revised 2026. arXiv:2508.08648.

[7] *Parabolic Kazhdan–Laumon and the Kloosterman Fourier Transform for Quadric Cones*.
2024; published in *Algebras and Representation Theory* (2026). arXiv:2410.17476.

Work in progress

- *Quantized Braverman–Kazhdan Reflection Functors*.
  With Nikolay Grantcharov. In preparation.
- *Geometrization of the Schrödinger Model for the Minimal Representation of the Even Orthogonal Group: the Étale Setting*. In preparation.

Awards and fellowships

2025–2026 **Lewis Blake Award for Excellence in Teaching**
Duke University.

2025 **Recognition for teaching excellence**, Trinity College, Duke University
Spring 2025 MATH 242 evaluations ranked in the top 5% of comparable Natural Sciences undergraduate courses in at least two evaluation categories.

2020 **Physical Sciences Teaching Prize**, University of Chicago

2018 **NSF Graduate Research Fellowship**

2018 **Phi Beta Kappa**, Alpha of Massachusetts

2017, 2018 **Derek Bok Certificate of Distinction in Teaching**, Harvard University

2017 **Harvard College Scholar**

2016 **Wendell Prize Scholarship finalist**, Harvard University

2015 **Detur Book Prize**, Harvard University

Research mentoring

Fall 2026 **Supervised reading course and graduate seminar**
Supervising Kai Wang in algebraic stacks and Bun_G; the course has developed into a private seminar for 5–6 early graduate students on algebraic geometry, stacks, and the moduli space of principal G-bundles.

2025 **Co-advisor, senior honors thesis of Marie-Hélène Tomé**
Duke University, with Jayce Getz.
Towards a Geometric Understanding of Boundary Terms in the Poisson Summation Formula for Quadric Cones.
Recipient of Duke's 2025 Excellence in Mathematical Research Award.

Professional service and research programs

Journal refereeing: *Selecta Mathematica*; *International Mathematics Research Notices* (IMRN).

2024–2027 **Organizer**, Duke Number Theory Seminar

2024–2025 **Participant**, Duke Research Scholars
The Poisson summation conjecture. Mentor: Jayce Getz.

2023–2024 **Co-founder and co-organizer**, BunG Seminar
University of Chicago. Founded with Nikolay Grantcharov in Winter 2023.

Teaching experience

Duke University

Course instructor, except where noted below.

Spring 2027	**Commutative Algebra and Algebraic Geometry**, MATH 602 Scheduled.
Fall 2026	**Representation Theory**, MATH 603 / PHYSICS 603 Current.
Spring 2026	**Introduction to Abstract Algebra**, MATH 401 / MATH 701
Fall 2025	**Number Theory**, MATH 305S
Spring 2025	**Discrete Mathematics and Proofs**, MATH 242 / COMPSCI 232
Fall 2024	**Multivariable Calculus for Economics**, MATH 202D Discussion leader.

University of Chicago

Fall 2021, 2023	**Calculus II**, MATH 15200. Instructor.
Spring 2023	**Paris Mathematics Study Abroad Program** Teaching assistant: Algebraic Curves (Ngô Bảo Châu), p-adic Harmonic Analysis (John Boller), Introduction to Dynamical Systems (Simion Filip).
Spring 2022	**Linear Algebra**. Instructor.
Winter 2022	**Calculus III**, MATH 15300. Instructor.
2019–2020	**Inquiry-Based Learning Calculus** College Fellow; course assistant for Sarah Ziesler.

Harvard University

2015–2017	**Discrete Mathematics**, MATH 152. Upper-level course assistant.

Selected conferences and schools attended

July 14– Aug. 1, 2025	Summer Research Institute in Algebraic Geometry. Colorado State University, Fort Collins, Colorado.
July 22–26, 2024	WARTHOG: Coherent–Constructible Equivalences in Local Geometric Langlands and Representation Theory. University of Oregon.
July 1–19, 2024	Second Simons Math Summer Workshop: Moduli. Simons Center for Geometry and Physics, Stony Brook University.
June 10–14, 2024	Los Angeles Workshop on Representations and Geometry (LAWRGe): Relative Langlands Duality. University of Southern California.
June 2024	Summer School and Workshop on Relative Langlands Duality. University of Minnesota.
Mar. 2–6, 2024	Arizona Winter School: Abelian Varieties. University of Arizona.
Mar. 7–11, 2020	Arizona Winter School: Nonabelian Chabauty. University of Arizona.

Selected talks

Research seminars and conference talks

2026	Number Theory Seminar, Duke University Scheduled.
Oct. 11, 2026	*Geometrization of the Minimal Representation of the Even Orthogonal Group* AMS Fall Southeastern Sectional Meeting, Kennesaw State University.
Oct. 4, 2026	*Geometrization of the Minimal Representation of the Even Orthogonal Group* AMS Fall Eastern Sectional Meeting, George Washington University.
Aug. 21, 2026	*Geometrization of the Minimal Representation of the Even Orthogonal Group* Geometric Methods in Representation Theory Seminar, University of North Carolina at Chapel Hill.
Oct. 9, 2024	*Normalized Intertwining Operators for General Parabolics over a Finite Field* Number Theory Seminar, Duke University.
Feb. 10, 2024	*Normalized Intertwining Operators for General Parabolics over a Finite Field* Junior Number Theory Days, Johns Hopkins University.
May 9, 2022	*An Overview of Geometric Class Field Theory* Junior Number Theory Seminar, University of Oxford.

Selected expository seminar lectures, University of Chicago

Mar. 27, 2024	*Shtukas and V. Lafforgue's Proof of Global Langlands for Function Fields* BunG Seminar.
Jan. 3, 2024	*An Overview of the Tamagawa Number Theorem for Function Fields* BunG Seminar.
Oct. 11, 2023	*The Geometric Satake Correspondence* BunG Seminar.
Oct. 6, 2023	*Normalized Intertwining Operators over Finite Fields* Ngô's Langlands Program Seminar.
Oct. 2 and 9, 2023	*On the Bernstein Center for p-adic Groups* Emerton's Categorical Langlands Seminar.
Mar. 1, 2023	*A Mosaic of Examples and Basic Properties of BunG* BunG Seminar.
Jan. 2023	*Braverman–Kazhdan Normalized Intertwining Operators* Ngô's Langlands Program Seminar.
Nov. 2019	*Deligne–Lusztig Theory* Introductory lecture, Ngô's seminar on character sheaves.

Languages and additional activities

Spanish (fluent). Classical music: piano, violin, viola, and composition.

Hasty Pudding Theatricals (2015–2018); Harvard–Radcliffe Gilbert and Sullivan Society.